

VocPub Profile - Specification

URI

<https://linked.data.gov.au/def/vocpub/spec>

Title

VocPub Profile - Specification Document

Definition

This document specifies the VocPub Profile. It is to be used to inform people about the requirements that need to be met by data claiming to conform to the profile.

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2020-06-14

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2026-02-25

Version IRI

<https://linked.data.gov.au/def/vocpub/spec/5.5>

Version Information

5.5 2026-02: Added Shapes for Reqs 2.3.10 - 2.3.19

5.4 2026-02: Small fixes for a number of Issues raised in the Issue Tracker

5.3 ValPub valid. All propertyShape instances have IRIs, are not Blank Nodes. All Shapes have names & messages. SHACL-UI elements moved to shaclui.ttl

5.2 Implement Shape 2.1.2+3 using qualified shape pattern as this is supported by pySHACL

5.1 remove targetNode -> ConceptScheme shape, use schema:, not sdo:

5.0 change IRI to linked.data.gov.au

4.10 move all warning from PropertyShapes to NodeShapes

4.9 Fixed RDF syntax errors

4.8 Fixed Warning/Violation error for PropertyShapes

4.7 Fixed modified date alternate path listing bug

4.6 Fixed schema:historyNote->skos:historyNote bug

4.5 Added suggested predicates of license & copyrightHolder

4.4 Fixed versions across multiple Resources

- 4.3 Improved validator error messages by using more named Property Shapes
- 4.2: Included CONSTRUCT-based pre-validation inference in validator. First Git tagged version
- 4.1: Added Requirements 2.1.10, 2.1.11 & 2.1.12 and example RDF
- 4.0: Added a SPARQL function to allow for the inferencing of `skos:inScheme` predicates, `skos:broader` / `skos:narrower` and `skos:topConceptOf` / `skos:hasTopConcept` pairs of inverse predicates
- 3.3: Converted validator metadata to schema.org, enabled bibliographic references for Concepts, enabled DCTERMS or schema.org for many ConceptScheme predicates; simplified 2.1.6 from two Requirements to one; included `skos:topConceptOf` in 2.1.8 for Concepts at the top of the hierarchy; collapsed title & definition requirement pairs to single requirements
- 3.2: Allowed `dcterms:provenance` and `skos:historyNote`; removed max restriction on `dcterms:source` & `prov:wasDerivedFrom`
- 3.1: Changed `dcterms:provenance` to `skos:historyNote`
- 3.0: Removed Requirement-2.3.5 (identifiers) as these are auto-generated in systems like VocPrez; Added Requirement-2.1.10 & 2.1.11 and sub parts to test for qualifiedDerivation and status of a ConceptScheme

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Publisher

Australian Government Linked Data Working Group

Further metadata

This specification is part of the *VocPub Profile*. See that profile's main document for License & Rights information and other metadata not given here.

Profile URI

<https://linked.data.gov.au/def/vocpub>

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Abstract

This is the specification document of the VocPub **profile** of **SKOS**. It defines the requirements that data must satisfy to be considered conformant with this profile.

This specification document cannot be used for testing conformance of RDF resources to this profile: that role belongs to the *validation* resource within this profile:

- <https://linked.data.gov.au/def/vocpub/validator>

For the list of all resources within this profile, see the profile definition:

- <https://linked.data.gov.au/def/vocpub>

Namespaces

This document refers to elements of various ontologies by short codes using namespace prefixes. The prefixes and their corresponding namespaces' URIs are:

dcat

<http://www.w3.org/ns/dcat#>

dcterms

<http://purl.org/dc/terms/>

isorole

<http://def.isotc211.org/common/RoleCode/>

prof

<http://www.w3.org/ns/dx/prof/>

prov

<http://www.w3.org/ns/prov#>

rdfs

<http://www.w3.org/2000/01/rdf-schema#>

reg

<http://purl.org/linked-data/registry#>>

schema

<https://schema.org/>

skos

<http://www.w3.org/2004/02/skos/core#>

time

<http://www.w3.org/2006/time#>

xsd

<http://www.w3.org/2001/XMLSchema#>

1. Introduction

Many organisations use the Simple Knowledge Organization System Reference (SKOS)^{ref} to represent vocabularies in a form that can be read by humans and consumed by machines; that is, in *Semantic Web* form^{ref}.

This profile defines a *vocabulary* as a controlled collection of defined terms - Concepts - that may or may not contain relationships between Concepts and relationships to Concepts in other vocabularies.

This document specifies a *profile* of SKOS and for profile, the definition of from the *Profiles Vocabulary*^{ref} is used. A *profile* is:

A specification that constrains, extends, combines, or provides guidance or explanation about the usage of other specifications.

Here, the *other specification* being profiled is SKOS.

In the next section, this document describes how SKOS's elements must be presented - in certain arrangements with respect to one another and with certain predicates to indicate properties - to make a vocabulary that conforms to this profile.

This specification's rules/requirements - are numbered and indicated in red text.

1.1 Data Expansion

Some SKOS elements - classes and predicates - can be inferred based on rules present in the SKOS model. For example, `skos:broader` and `skos:narrower` are inverse predicates thus if I have `<A> skos:broader <B>`, I can infer `<B> skos:narrower <A>`.

This profile allows data to be supplied in a minimalist form that is not conformant to this specification until a series of calculations, based on certain SKOS rules, are carried out on it through a process known as *data expansion*.

The particular rules that may be applied to data before validation with this profile's validator are given in the table below. These rules are enacted by the application of a series of SPARQL^{ref} queries to the data, all of which are packaged up inside a SHACL^{ref} file in this profile's repository.

Rule	Description	SPARQL Query
hasTopConcept	Calculate <code>skos:hasTopConcept</code> as the inverse to <code>skos:topConceptOf</code>	<pre>CONSTRUCT { ?cs skos:hasTopConcept ?c . } WHERE { ?c skos:topConceptOf ?cs . }</pre>

topConceptOf	Calculate skos:topConceptOf as the inverse to skos:hasTopConcept	<pre> CONSTRUCT { ?c skos:topConceptOf ?cs . } WHERE { ?cs skos:hasTopConcept ?c . } </pre>
broader	Calculate skos:broader as the inverse to skos:narrower	<pre> CONSTRUCT { ?c skos:broader ?b . } WHERE { ?b skos:narrower ?c . } </pre>
narrower	Calculate skos:narrower as the inverse to skos:broader	<pre> CONSTRUCT { ?c skos:narrower ?n . } WHERE { ?n skos:broader ?c . } </pre>
inScheme	Calculate skos:inScheme for all Concepts, based on their linking to a Concept Scheme via skos:broader / skos:topConceptOf property path	<pre> CONSTRUCT { ?c skos:inScheme ?cs . } WHERE { ?c skos:broader*/skos:topConceptOf ?cs . } </pre>
Concept provenance	Infer provenance predicates for a Concept when they don't have their own but their containing Concept Scheme does	<pre> CONSTRUCT { ?c ?p ?o3 } WHERE { ?c skos:inScheme ?cs . VALUES ?p { prov:wasDerivedFrom skos:historyNote schema:citation dcterms:source dcterms:provenance } ?cs ?p ?o . OPTIONAL { ?c ?p ?o2 . } } </pre>

		<code>BIND (COALESCE(?o2, ?o) AS ?o3)</code> }
--	--	---

Application of these rules will allow a Concept supplied without any provenance predicates to be calculated from its containing Concept Scheme, thus satisfying Requirement 2.3.4.

The SHACL file containing all the expansion queries is available at:

- <https://linked.data.gov.au/def/vocpub/expander>

A Python script able to execute the expansion rule on RDF data before validation is available at:

- <https://linked.data.gov.au/def/vocpub/validate>

1.2 Dependencies

To characterise vocabularies according to the mandatory and suggested Requirements of this Profile, several other vocabularies will need to be used. This profile is therefore dependent on those vocabularies. For this reason, copies of these vocabularies are maintained within the repository of this profile and are accessible as per the details table below.

Vocabulary	Description	Where used	Local Copy
ISO 19115 Role Codes	Codes indicating the role Agents - People and Organisations - play in relation to data	For the predicate <code>prov:hadRole</code> , applied to an <code>Attribution</code> indicated by a <code>prov:qualifiedAttribution</code> predicate for a Concept Scheme	https://def.isotc211.org/common/RoleCode
Registry Statuses	The registration statuses of items in a Register, as per ISO19135	For the suggested vocabulary predicate <code>schema:status</code> (preferred) or <code>reg:status</code> , as per Requirement 2.1.10 & 2.3.5	https://linked.data.gov.au/def/reg-statuses
Vocab Derivation Modes	The modes by which one vocabulary may derive from another	For the suggested vocabulary predicate <code>prov:qualifiedDerivation</code> , as per Requirement 2.1.11	https://linked.data.gov.au/def/vocdermods

2. Elements & Requirements

2.1 Vocabulary

This profile identifies Semantic Web objects with URI-based persistent identifiers. For this reason:

2.1.01 Each vocabulary *MUST* be identified by a IRI

As per the *SKOS Primer*^{ref}, a document guiding the use of SKOS:

*concepts usually come in carefully
compiled vocabularies, such as thesauri
or classification schemes. SKOS offers
the means of representing such KOSs
using the `skos:ConceptScheme` class.*

For this reason, this profile requires that:

2.1.02 Each vocabulary *MUST* be presented as a single Concept Scheme object

For ease of data management:

2.1.03 Each vocabulary *MUST* be presented in a single RDF file which does not contain information other than that which is directly part of the vocabulary

To ensure vocabularies can be catalogued effectively and governed:

2.1.04 Each vocabulary *MUST* have exactly one title and at least one definition indicated using the `skos:prefLabel` and the `skos:definition` predicates respectively that must give textual literal values. Only one definition per language is allowed

NOTE: Unlike the general directions for the use of SKOS (the *SKOS "Primer"*) labels in multiple languages should be indicated with `skos:altLabel` predicates, not all with `skos:prefLabel`, i.e. there should only ever be one `skos:prefLabel` value. If multiple definitions are given, the one in the language of the label is considered primary.

2.1.05 Each vocabulary *MUST* have exactly one created date and exactly one modified date indicated using the `schema:dateCreated` and `schema:dateModified` or `dcterms:created` and `dcterms:modified` predicates respectively that must be either an `xsd:date`, `xsd:dateTime` or `xsd:dateTimeStamp` literal value

2.1.06 Each vocabulary *MUST* have at least one creator, indicated using `schema:creator` or `dcterms:creator` predicate and exactly one publisher, indicated using `schema:publisher` or `dcterms:publisher`, all of which *MUST* be IRIs indicating instances of `schema:Person`, or `schema:Organization`. A `prov:qualifiedAttribution` predicate indicating an Agent with the `prov:hadRole` predicate indicating the value `isole:originator` or `isole:publisher` may be used instead of `schema:creator` & `schema:publisher`, respectively

To be able to link SKOS vocabularies to their non-vocabulary source information:

2.1.07 The origins of a Concept Scheme *MUST* be indicated by at least one of the following predicates: `skos:historyNote`, `schema:citation`, `prov:wasDerivedFrom`. `dcterms:source` *MAY* be used instead of `schema:citation` and `dcterms:provenance` *MAY* be used instead of `skos:historyNote` but the `schema.org` and SKOS predicates are preferred.

If a vocabulary is based on another Semantic Web resource, such as an ontology or another vocabulary, `prov:wasDerivedFrom` should be used to indicate that resource's IRI. If the vocabulary is based on a resource that is identified by a IRI but which is not a Semantic Web resource, `schema:citation` should be used to indicate the resource's IRI with the `xsd:anyURI` datatype. If the vocabulary is based on something which cannot be identified by IRI, a statement about the vocabulary's origins should be given in a literal value indicated with the `skos:historyNote` predicate. If the vocabulary is not based on any other resource or source of information, i.e. this vocabulary is its only expression, this should be communicated by use of the `skos:historyNote` indicating the phrase "This vocabulary is expressed for the first time here".

The use of `dcterms:source` & `dcterms:provenance` is to maintain compatability with previous versions of VocPub only and may eventually be disallowed.

To ensure that all the terms within a vocabulary are linked to the main vocabulary object, the Concept Scheme:

2.1.08 All Concept instances within a Concept Scheme *MUST* be contained in a single term hierarchy using `skos:hasTopConcept` / `skos:topConceptOf` predicates indicating the broadest Concepts in the vocabulary and then `skos:broader` and/or `skos:narrower` predicates for all non-broadest Concepts in a hierarchy that contains no cycles.

To unambiguously link the term hierarchy within a vocabulary to the vocabulary itself:

2.1.09 Each vocabulary's Concept Scheme *MUST* link to at least one Concept within the vocabulary with the `skos:hasTopConcept` predicate

To communicate the *Registry Status* of the vocabulary:

2.1.10 The status of the vocabulary as a whole, according to the Registry Status standard^{ref}, *SHOULD* be given with the predicate `schema:status` (preferred) or `reg:status` indicating a Concept from the *Registry Statuses* vocabulary (<https://linked.data.gov.au/def/reg-statuses>).

To indicate whether and if so how this vocabulary has been derived from another vocabulary:

2.1.11 The derivation status of the vocabulary *SHOULD* be given with the predicate `prov:qualifiedDerivation` indicating a Blank Node that contains the predicated `prov:entity`, to indicate the vocabulary derived from and `prov:hadRole` to indicate the mode of derivation which *SHOULD* be taken from the *Vocabulary Derivation Modes* vocabulary (<https://linked.data.gov.au/def/vocdermods>).

Example data for a vocabulary indicating that it is an extension to another vocabulary using the mechanism defined in Requirement 2.1.2 is:

```
# Vocab X is derived from Vocab Y and is an extension of it
<http://example.com/vocab/x>
```

```

    a skos:ConceptScheme ;
    skos:prefLabel "Vocab X"@en ;
    ...
    prov:qualifiedDerivation [
        prov:entity <http://example.com/vocab/y> ; # Vocab Y
        prov:hadRole <https://linked.data.gov.au/def/vocdermods/extension> ;
    ] ;
.

```

To high-level theming of a vocabulary:

2.1.12 High-level theming of a vocabulary *SHOULD* be given using the `schema:keywords` predicate indicating Concepts from another vocabulary. Alternatively, `dc:theme` *MAY* be used. Text literal values for either predicate *SHOULD NOT* be used.

To indicate license, copyright:

2.1.13 Any licence pertaining to the reuse of a vocabulary's content *SHOULD* be given using the `schema:license` predicate preferentially indicating the IRI of a license if in RDF form or a literal URL (datatype `xsd:anyURI`) if online but not in RDF form. If the licence is expressed in text, a literal text field may be indicated.

2.1.14 The copyright holder for the vocabulary *SHOULD* be given using the `schema:copyrightHolder` predicate preferentially indicating the IRI of an Agent or a Blank Node instance of an Agent containing details as per Agent requirements. A `prov:qualifiedAttribution` predicate indicating an Agent with the `prov:hadRole` predicate indicating the value `isole:rightsHolder` may be used instead of `schema:copyrightHolder`.

2.2 Collection

From the SKOS Primer^{ref}:

SKOS makes it possible to define meaningful groupings or "collections" of concepts. Collections may contain Concepts defined in any vocabulary, not just the one the Collection itself is defined in.

To ensure that Collection instances are identifiable and their meaning isn't obscure or lost:

2.2.01 Each Collection *MUST* have exactly one title and at least one definition indicated using the `skos:prefLabel` and the `skos:definition` predicates respectively that must give textual literal values. Only one definition per language is allowed

NOTE: Unlike the general directions for the use of SKOS (the **SKOS "Primer"**) labels in multiple languages should be indicated with `skos:altLabel` predicates,

not all with `skos:prefLabel`, i.e. there should only ever be one `skos:prefLabel` value. If multiple definitions are given, the one in the language of the label is considered primary.

If a Collection's grouping of Concepts is derived from an existing resource that is different from the ConceptScheme it is defined within:

2.2.02 The origins of a Collection, if different from its containing Concept Scheme, *SHOULD* be indicated by at least one of the following predicates: `skos:historyNote`, `schema:citation`, `prov:wasDerivedFrom`. `dcterms:source` *MAY* be used instead of `schema:citation` and `dcterms:provenance` *MAY* be used instead of `skos:historyNote` but the `schema.org` and SKOS predicates are preferred.

For compatability with previous versions of this Specification, `dcterms:provenance` MAY be used instead of `skos:historyNote` but the latter is the preferred predicate.

To help list Collections within vocabularies:

2.2.03 A Collection exists within a vocabulary *SHOULD* indicate that it is within the vocabulary by use of the `skos:inScheme` predicate. If it is defined for the first time in the vocabulary, it should also indicate this with the `rdfs:isDefinedBy` predicate

To ensure that a Collection isn't empty:

2.2.04 A Collection *MUST* indicate at least one Concept instance that is within the collection with use of the `skos:member` predicate. The Concept need not be defined by the Concept Scheme that defines the Collection

2.3 Concept

From the SKOS Primer^{ref}:

The fundamental element of the SKOS vocabulary is the concept. Concepts are the units of thought — ideas, meanings, or (categories of) objects and events—which underlie many knowledge organization systems

Vocabularies conforming to this profile must present at least one Concept within the vocabulary file and, as per requirements in Section 2.1, at least once Concept must be indicated as the top concept of the vocabulary.

To ensure that Concept instances are identifiable and their meaning isn't obscure or lost:

2.3.01 Each Concept *MUST* have exactly one title and at least one definition indicated using the `skos:prefLabel` and the `skos:definition` predicates respectively that must give textual literal values. Only one definition per language is allowed

NOTE: Unlike the general directions for the use of SKOS (the [SKOS "Primer"](#)) labels in multiple languages should be indicated with `skos:altLabel` predicates, not all with `skos:prefLabel`, i.e. there should only ever be one `skos:prefLabel` value. If multiple definitions are given, the one in the language of the label is considered primary.

To ensure that every Concept is linked to the vocabulary that defines it:

2.3.02 Each Concept in a vocabulary *MAY* indicate the vocabulary that defines it by use of the `rdfs:isDefinedBy` predicate indicating a Concept Scheme instance. If no such predicate is given, the Concept Scheme in the file that a Concept is provided in is understood to be the defining Concept Scheme

Note that the vocabulary that defines a Concept does not have to be the vocabulary in the file being validated. This is to allow for Concept instance reuse across multiple vocabularies.

Since a Concept may be used in more than one vocabulary:

2.3.03 Each Concept in a vocabulary *MUST* indicate that it appears within that vocabulary's hierarchy of Concepts either directly by use of the `skos:topConceptOf` predicate indicating the vocabulary or indirectly by use of one or more `skos:broader` / `skos:narrower` predicates placing the Concept within a chain of other Concepts, the top concept of which uses the `skos:topConceptOf` predicate to indicate the vocabulary.

If a Concept is derived from an existing resource and that derivation is not already covered by source information for the vocabulary that it is within:

2.3.04 The origins of a Concept, if different from its containing Concept Scheme, *SHOULD* be indicated by at least one of the following predicates: `skos:historyNote`, `schema:citation`, `dcterms:source` or `prov:wasDerivedFrom` or `dcterms:provenance`.

If a Concept is based on another Semantic Web resource, such as another Concept or other defined object, `prov:wasDerivedFrom` should be used to indicate that resource's IRI. If the Concept is based on a resource that is identified by a IRI but which is not a Semantic Web resource, `dcterms:source` should be used to indicate the resource's IRI. If the vocabulary is based on something which cannot be identified by IRI, a statement about the vocabulary's origins should be given in a literal value indicated with the `skos:historyNote` predicate. If the vocabulary is not based on any other resource or source of information, i.e. this vocabulary is its only expression, this should be communicated by use of the `skos:historyNote` indicating the phrase "This vocabulary is expressed for the first time here".

To communicate the *Registry Status* of the vocabulary:

2.3.05 The status of a Concept, according to the Registry Status standard^{ref}, *MAY* be given with the predicate `schema:status` (preferred) or `reg:status` indicating a Concept from the *Registry Statuses* vocabulary (<https://linked.data.gov.au/def/reg-statuses>).

To link a Concept to other things with literal identifiers:

2.3.10 If a Concept indicates a `skos:notation`, its range value **MUST** be a literal. Note that it **SHOULD** be of a custom datatype

The use of custom datatypes with SKOS `notation` predicates is explained in [SKOS Primer, Section 4.6 Notations](#).

To describe the history of a Concept (it's development):

2.3.11 If a Concept indicates a `skos:historyNote`, its range value **MUST** be a literal. A plain string value (`xsd:string`) may be used or, if values in multiple languages are desired, then multiple language string literals (`rdf:langString`) may be used, but only one per language and plain strings and language strings should not be mixed

To indicate another Concept from which a Concept was derived:

2.3.12 If a Concept indicates `was derived from`, its range value **MUST** be an IRI

To indicate another Concept which a Concept is a revision of, revision of being a closer derivation than the more generic was derived from, as above:

2.3.13 If a Concept indicates `revision of`, its range value **MUST** be an IRI

To indicate another Concept that a Concept replaces:

2.3.14 If a Concept indicates `replaces`, its range value **MUST** be an IRI

To indicate another Concept that a Concept is replaced by, the inverse of `replaces` above:

2.3.15 If a Concept indicates `is replaced by`, its range value **MUST** be an IRI

To indicate a version identifier for a Concept:

2.3.16 If a Concept indicates `version`, its range value **MUST** be a plain string (`xsd:string`)

To indicate details about when a Concept was created and modified but not when it is applicable for:

2.3.17 If a Concept indicates `a date created`, its range value **MUST** be a date (`xsd:date`)

2.3.18 If a Concept indicates `a date modified`, its range value **MUST** be a date (`xsd:date`)

To indicate the period of time that a Concept is relevant for:

2.3.19 If a Concept indicates `a temporal coverage`, its range value **MUST** be a temporal entity (`time:TemporalEntity`) with either or both `schema:startDate` and `schema:endDate` values which must be dates (`xsd:date`)

2.4 Agent

To be consistent with other Semantic Web representations of Agents, vocabularies' associated Agents, creator & publisher must be certain typed RDF values:

2.4.1 Each Agent associated with a vocabulary *MUST* be typed as an `schema:Person` or `schema:Organization`

To ensure human readability and association of Agents with their non-Semantic Web (real world) form:

2.4.2 Each Agent *MUST* give either exactly one name with the `schema:name` predicate indicating a plain string text value or at least one `schema:name` predicate indicating a language string (e.g. "Nicholas"@en). If multiple language strings are used, the languages must be unique.

To ensure that Agents are linked to non-Semantic Web forms of identification:

2.4.3 Each Agent *MUST* indicate either a `schema:url` (for organizations) or a `schema:email` (for people) predicate with a URL or email value

To link to Agent registers using non-Semantic Web identifiers for Agents:

2.4.4 Each Agent *SHOULD* indicate any non-Semantic Web identifiers for Agents with the `schema:identifier` predicate with literal identifier values, preferentially with custom data types that define the form of the identifier.

NOTE: This method of providing identifiers with specialised datatypes is the same as that specified for `skos:notation` values in the *SKOS Primer*.

3. References

PROF

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